INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 13

1. Experiment Title:

UNION and INTERSECT

2. Aim:

Combine names from two clubs, find common names, and include duplicate names using set operators.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

UNION combines result sets and removes duplicates. INTERSECT returns rows common to both result sets. UNION ALL combines result sets while retaining duplicates. The SELECT statements must return compatible columns.

5. Algorithm / Procedure:

1. Create club_a and club_b and insert the supplied names.
2. Use UNION and sort the unique combined names.
3. Use INTERSECT to find names in both clubs.
4. Use UNION ALL to keep repeated names and sort the result.

6. Program / Queries / Commands:

```
CREATE TABLE club_a (
    student_name VARCHAR(30)
);

INSERT INTO club_a (student_name)
VALUES ('Asha'), ('Rohan'), ('Neha');

CREATE TABLE club_b (
    student_name VARCHAR(30)
);

INSERT INTO club_b (student_name)
VALUES ('Neha'), ('Pooja'), ('Meera');

-- Task 1
SELECT student_name FROM club_a
UNION
SELECT student_name FROM club_b
ORDER BY student_name;

-- Task 2
SELECT student_name FROM club_a
INTERSECT
SELECT student_name FROM club_b
ORDER BY student_name;

-- Task 3
SELECT student_name FROM club_a
UNION ALL
SELECT student_name FROM club_b
ORDER BY student_name;
```

7. Input:

club_a: Asha, Rohan, Neha. club_b: Neha, Pooja, Meera.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: student_name
Asha
Meera
Neha
Pooja
Rohan
Task 2: student_name
Neha
Task 3: student_name
Asha
Meera
Neha
Neha
Pooja
Rohan

9. Result:

The SQL statements for Experiment 13 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is UNION used for?
2. How is UNION different from UNION ALL?
3. What does INTERSECT return?
4. What condition must the SELECT statements meet when using set operators?